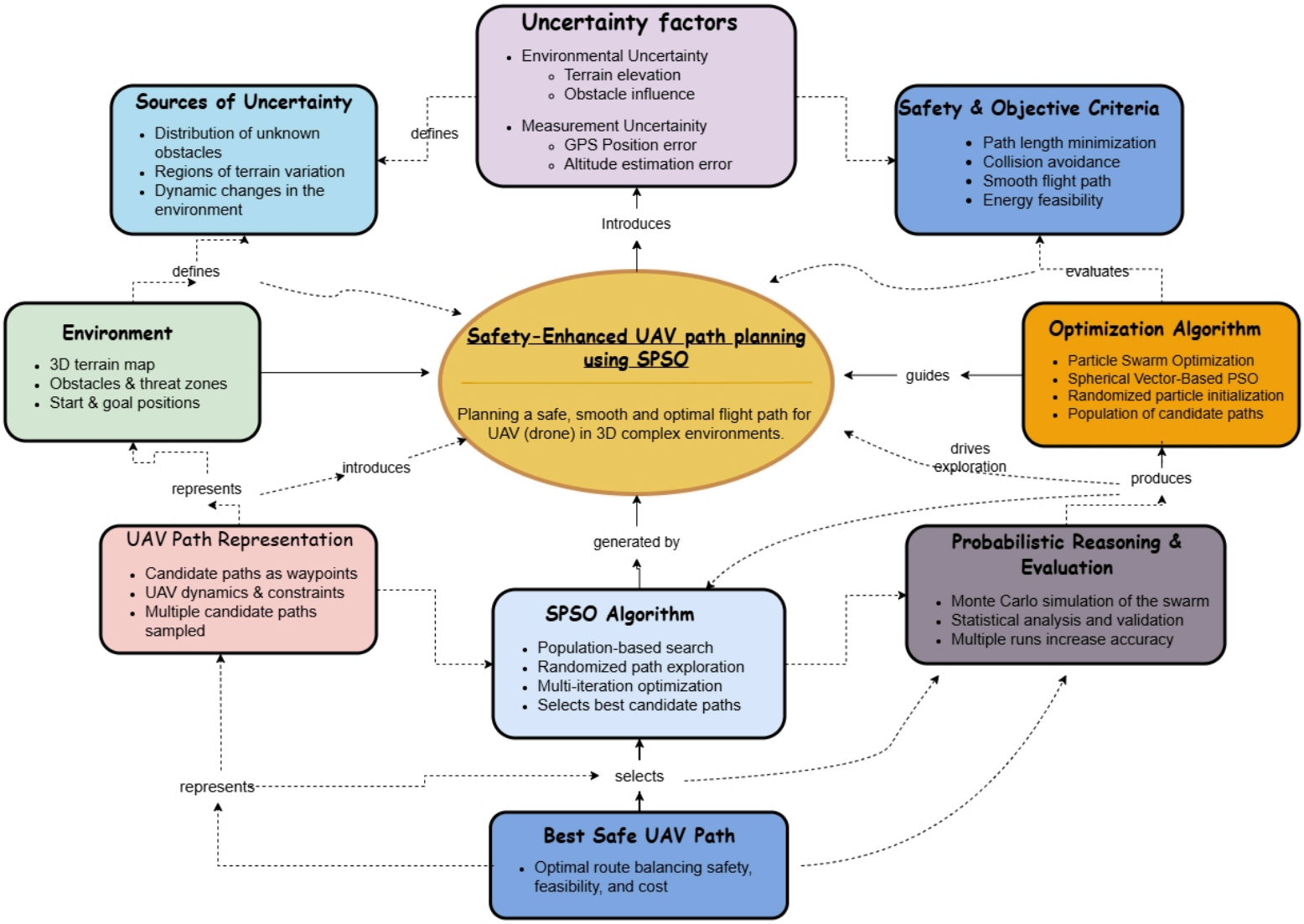
Uncertainty factors
Environmental Uncertainty
Terrain elevation
Obstacle influence
Measurement Uncertainity
GPS Position error
Altitude estimation error
Sources of Uncertainty
Distribution of unknown obstacles
Regions of terrain variation
Dynamic changes in the environment
defines
Safety & Objective Criteria
Path length minimization
Collision avoidance
Smooth flight path
Energy feasibility
Introduces
defines
evaluates
Environment
3D terrain map
Obstacles & threat zones
Start & goal positions
Safety-Enhanced UAV path planning using SPSO
Planning a safe, smooth and optimal flight path for UAV (drone) in 3D complex environments.
guides
Optimization Algorithm
Particle Swarm Optimization
Spherical Vector-Based PSO
Randomized particle initialization
Population of candidate paths
drives exploration
produces
represents
introduces
generated by
UAV Path Representation
Candidate paths as waypoints
UAV dynamics & constraints
Multiple candidate paths sampled
SPSO Algorithm
Population-based search
Randomized path exploration
Multi-iteration optimization
Selects best candidate paths
Probabilistic Reasoning & Evaluation
Monte Carlo simulation of the swarm
Statistical analysis and validation
Multiple runs increase accuracy
represents
selects
Best Safe UAV Path
Optimal route balancing safety, feasibility, and cost